

Jaundel Simon

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EDUCATION

Toronto Metropolitan University

Bachelor of Engineering in Computer Engineering

Toronto, ON

Sep 2023 – May 2028

- **Relevant Coursework:** Computer Architecture, Digital Systems, Data Structures & Algorithms, Microprocessor Systems, Object-Oriented Design

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C, VHDL, MATLAB

Machine Learning & Simulation: NumPy, PyTorch, TensorFlow.js, SciPy, scikit-learn, Three.js, WebGL 2

Hardware & Tools: FPGA/RTL, GHDL, Intel Quartus Prime, ModelSim/Questa, Git, GitHub Actions, Vite

PROJECTS

RISC-NPU: ISA-Integrated Neural Acceleration | VHDL, GHDL, Quartus Prime

Sep 2026

- Designed a custom 32-bit RISC CPU with signed INT8 MAC/ReLU instructions, cutting a 64-element dot product from 12,648 to 778 cycles (16.26× kernel speedup; 6.12× including initialization) in cycle-accurate simulation
- Implemented iterative and parallel multiplier variants with fused retirement and 32-bit saturating accumulation; closed 12 matched Cyclone IV fits under a 50 MHz constraint
- Built a 13-test GHDL regression and reproducible experiment pipeline covering 592 kernel runs, 1,296 output checks, 196,608 signed-byte products, and 12,294 conversions

Morphic: Editable Physics from Captured Objects | JavaScript, Three.js, WebGL 2, Python

Sep 2026

- Built a browser studio that converts textured GLBs into interactive tetrahedral XPBD simulations with a 120 Hz worker, GPU skinning, collision handling, and portable deterministic replay
- Raised a 75,649-vertex tablet workload from 10.54 to 59.96 median FPS by moving deformation to the GPU; verified zero same-runtime replay drift across 12 600-step trajectories and 13 automated tests
- Implemented observation-based response fitting and matching Python/JavaScript solvers with zero coordinate difference across four frozen PhysTwin trials while preserving held-out failure evidence

MedMNIST from Scratch | Python, NumPy, PyTorch, scikit-learn

Sep 2026

- Generalized an 80-line NumPy MLP with handwritten forward/backpropagation to four 28×28 medical-image benchmarks, reaching 0.825 accuracy and 0.931 AUC on PneumoniaMNIST
- Implemented balanced metrics, saliency, occlusion, FGSM, cross-view shift, 5-client FedAvg, Forward-Forward, and pruning; retained 0.843 pneumonia accuracy after removing 90% of weights and fine-tuning

SEVEN: Energy-Aware LLM Routing | Python, Lemonade Server, OpenAI/Groq APIs

Nov 2025

- Founded and led a 5-person team to build a router across local SLM, direct API, and cloud paths; won TMU Hack the World and a \$1,625 prize
- Integrated Lemonade Server with runtime CPU/GPU/NPU selection, exponential-backoff retries, local/cloud fallback, and per-query energy telemetry; modeled a 38% reduction under a 40% local-routing scenario
- Authored setup runbooks and managed pull requests and code reviews across the terminal UI, energy profiling, and SQL-backed conversation logging

EXPERIENCE

Graphic Installer

Signscapers Installation Inc.

Sep 2023 – Aug 2026

Greater Toronto Area

- Installed wayfinding signs, branded wall graphics, privacy-frost vinyl, and point-of-purchase displays across client sites including the NBA, Scotiabank, TIFF, Microsoft, RBC, and Wine Rack
- Coordinated with client representatives and installation crews to confirm placement requirements, communicate site constraints, and address on-site questions throughout each installation